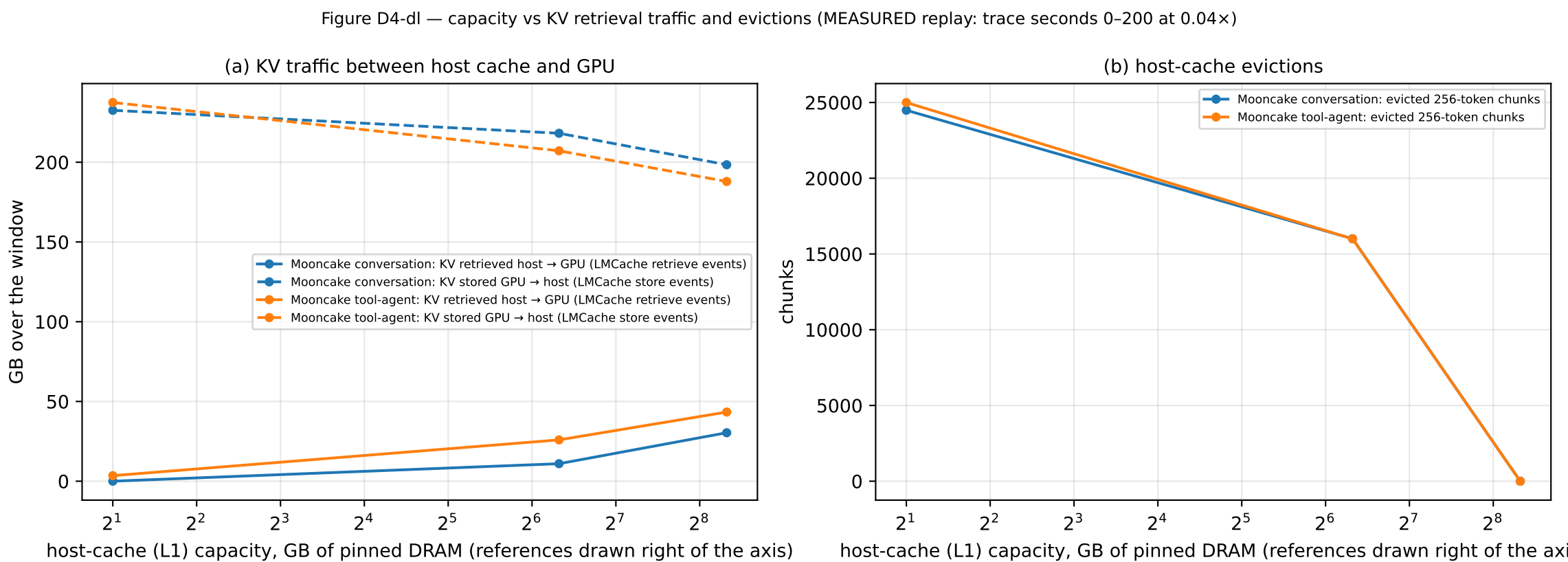
Figure D4-dl — capacity vs KV retrieval traffic and evictions (MEASURED replay: trace seconds 0–200 at 0.04×)
(a) KV traffic between host cache and GPU
GB over the window
0
50
100
150
200
Mooncake conversation: KV retrieved host → GPU (LMCache retrieve events)
Mooncake conversation: KV stored GPU → host (LMCache store events)
Mooncake tool-agent: KV retrieved host → GPU (LMCache retrieve events)
Mooncake tool-agent: KV stored GPU → host (LMCache store events)
2^1 2^2 2^3 2^4 2^5 2^6 2^7 2^8
host-cache (L1) capacity, GB of pinned DRAM (references drawn right of the axis)
(b) host-cache evictions
chunks
0
5000
10000
15000
20000
25000
Mooncake conversation: evicted 256-token chunks
Mooncake tool-agent: evicted 256-token chunks
2^1 2^2 2^3 2^4 2^5 2^6 2^7 2^8
host-cache (L1) capacity, GB of pinned DRAM (references drawn right of the axi